

Math 142, F20
Exam 1

Name: *Answer*

Instructions:

- There are totally 5 problems.
- You must **show all of your work to receive a credit for a correct answer.**
- No calculator is allowed in the exam.

Problem	Points	Score
1	26	
2	20	
3	20	
4	20	
5	20	
Total	106	

Good Luck !

Problem 1. (24 points) Evaluate the following integrals using appropriate substitutions.

(a) $\int_0^{\pi/2} \sin x \sqrt{1 - \cos x} \, dx$

let $u = 1 - \cos x$, $du = \sin x \, dx$

when $x=0$, $u = 1 - \cos 0 = 1 - 1 = 0$

$x = \frac{\pi}{2}$, $u = 1 - \cos(\frac{\pi}{2}) = 1 - 0 = 1$

Thus

$$\int_0^{\pi/2} \sin x \sqrt{1 - \cos x} \, dx = \int_0^1 \sqrt{u} \, du$$

$$= \left[\frac{2}{3} u^{3/2} \right]_{u=0}^{u=1}$$

$$= \frac{2}{3} (1^{3/2} - 0^{3/2}) = \frac{2}{3}$$

(b) $\int x(x-1)^{10} \, dx$

let $u = x-1$, $du = dx$

$x = u+1$

$\int x(x-1)^{10} \, dx = \int (u+1) u^{10} \, du$

$= \int u^{11} + u^{10} \, du$

$= \frac{u^{12}}{12} + \frac{u^{11}}{11} + C$

$= \frac{(x-1)^{12}}{12} + \frac{(x-1)^{11}}{11} + C$

(c) $\int \frac{x}{x^4 + 1} \, dx$

let $u = x^2$, $du = 2x \, dx$

$\int \frac{x}{x^4 + 1} \, dx = \frac{1}{2} \int \frac{1}{u^2 + 1} \, du$

$= \frac{1}{2} \arctan(u) + C$

$= \frac{1}{2} \arctan(x^2) + C$

(d) $\int_0^{\pi/4} \tan^2 \theta \sec^4 \theta \, d\theta$

let $u = \tan \theta$, $du = \sec^2 \theta \, d\theta$

when $\theta = \pi/4$, $u = \tan \pi/4 = 1$

$\theta = 0$, $u = \tan 0 = 0$

$\int_0^{\pi/4} \tan^2 \theta \sec^4 \theta \, d\theta$

$= \int_0^{\pi/4} \tan^2 \theta (1 + \tan^2 \theta) \sec^2 \theta \, d\theta$

$= \int_0^1 u^2 (1 + u^2) \, du$

$= \int_0^1 u^4 + u^2 \, du$

$= \left[\frac{u^5}{5} + \frac{u^3}{3} \right]_{u=0}^{u=1} = \frac{1}{5} + \frac{1}{3} = \frac{8}{15}$

Problem 2. (20 points) Evaluate the following integrals using integration by parts.

(a) $\int_0^1 x e^{-3x} dx$

let $u = x$, $dv = e^{-3x} dx$

$du = dx$, $v = -\frac{1}{3}e^{-3x}$

$$\int_0^1 x e^{-3x} dx = x \cdot \left(-\frac{1}{3}e^{-3x}\right) \Big|_{x=0}^{x=1} - \int_0^1 \left(-\frac{1}{3}e^{-3x}\right) dx$$

$$= -\frac{1}{3}e^{-3} + \int_0^1 \frac{1}{3}e^{-3x} dx$$

$$= -\frac{1}{3}e^{-3} + \left[-\frac{1}{9}e^{-3x}\right]_{x=0}^{x=1}$$

$$= -\frac{1}{3}e^{-3} - \frac{1}{9}e^{-3} + \frac{1}{9}$$

$$= \underline{\underline{-\frac{4}{9}e^{-3} + \frac{1}{9}}}$$

(b) $\int \sin \sqrt{t} dt$

(Hint: first use the substitution $x = \sqrt{t}$)

let $x = \sqrt{t}$, then $x^2 = t$, $2x dx = dt$

$$\int \sin \sqrt{t} dt = \int (\sin x)(2x dx)$$

$$= 2 \int x \sin x dx \leftarrow$$

let $u = x$, $dv = \sin x dx$
 $du = dx$, $v = -\cos x$

$$= 2 \left[-x \cos x - \int (-\cos x) dx \right]$$

$$= -2x \cos x + 2 \int \cos x dx$$

$$= -2x \cos x + 2 \sin x + C$$

$$= \underline{\underline{-2\sqrt{t} \cos \sqrt{t} + 2 \sin \sqrt{t} + C}}$$

Problem 3. (20 points)

(a) Evaluate $\int \sqrt{4-t^2} dt$ using appropriate trigonometric substitution.

let $t = 2 \sin \theta$, then $dt = 2 \cos \theta d\theta$ and $\sqrt{4-t^2} = 2 \cos \theta$.

$$\downarrow$$

$$\theta = \arcsin\left(\frac{t}{2}\right)$$



$$\sin \theta = \frac{t}{2}$$

$$\cos \theta = \frac{\sqrt{4-t^2}}{2}$$

$$\begin{aligned} \int \sqrt{4-t^2} dt &= \int 2 \cos \theta \cdot 2 \cos \theta d\theta \\ &= \int 4 \cos^2 \theta d\theta = \int 4 \cdot \frac{1 + \cos 2\theta}{2} d\theta \\ &= 2\theta + \sin 2\theta + C \\ &= 2\theta + 2 \sin \theta \cos \theta + C \\ &= 2 \arcsin\left(\frac{t}{2}\right) + 2 \cdot \frac{t}{2} \cdot \frac{\sqrt{4-t^2}}{2} + C \\ &= 2 \arcsin\left(\frac{t}{2}\right) + \frac{t\sqrt{4-t^2}}{2} + C \end{aligned}$$

(b) Evaluate the improper integral or show that it is divergent: $\int_2^{+\infty} \frac{1}{x(\ln x)^2} dx$

$$\int_2^{+\infty} \frac{1}{x(\ln x)^2} dx = \lim_{b \rightarrow +\infty} \int_2^b \frac{1}{x(\ln x)^2} dx$$

$$\int \frac{1}{x(\ln x)^2} dx \xrightarrow[u = \ln x]{du = \frac{1}{x} dx} \int \frac{1}{u^2} du = -\frac{1}{u} + C = -\frac{1}{\ln x} + C$$

$$\begin{aligned} \Rightarrow \lim_{b \rightarrow +\infty} \int_2^b \frac{1}{x(\ln x)^2} dx &= \lim_{b \rightarrow +\infty} \left[-\frac{1}{\ln x} \right]_{x=2}^{x=b} \\ &= \lim_{b \rightarrow +\infty} -\frac{1}{\ln b} + \frac{1}{\ln 2} = \frac{1}{\ln 2} \end{aligned}$$

Convergent!

Problem 4. (20 points) Evaluate the following integrals of rational functions.

(a) $\int \frac{x-12}{x^2+x-6} dx$

$$x^2+x-6 = (x+3)(x-2) \rightarrow x=2 \text{ or } x=-3, \text{ two roots.}$$

$$\frac{x-12}{x^2+x-6} = \frac{x-12}{(x+3)(x-2)} = \frac{A}{x+3} + \frac{B}{x-2} = \frac{A(x-2)+B(x+3)}{(x+3)(x-2)}$$

$$\Rightarrow x-12 = A(x-2) + B(x+3)$$

$$\text{let } x=2 \Rightarrow -10 = 5B \Rightarrow B=-2$$

$$\text{let } x=-3 \Rightarrow -15 = -5A \Rightarrow A=3$$

$$\begin{aligned} \Rightarrow \int \frac{x-12}{x^2+x-6} dx &= \int \frac{3}{x+3} - \frac{2}{x-2} dx \\ &= \underline{3 \ln|x+3| - 2 \ln|x-2| + C} \end{aligned}$$

(b) $\int \frac{2x^2-2x+1}{x^4+x^2} dx$

$$x^4+x^2 = x^2(x^2+1)$$

$$\Rightarrow \frac{2x^2-2x+1}{x^4+x^2} = \frac{2x^2-2x+1}{x^2(x^2+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1}$$

$$= \frac{A x(x^2+1) + B(x^2+1) + (Cx+D)x^2}{x^2(x^2+1)} = \frac{Ax^3 + Ax + Bx^2 + B + Cx^3 + Dx^2}{x^2(x^2+1)}$$

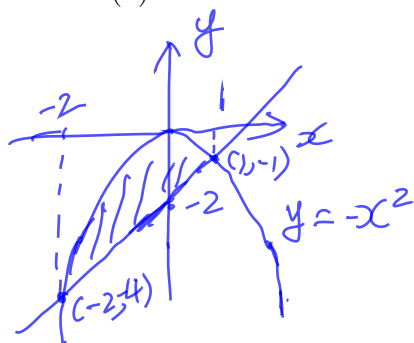
$$= \frac{(A+C)x^3 + (B+D)x^2 + Ax + B}{x^2(x^2+1)}$$

$$\Rightarrow \begin{cases} A+C=0 \\ B+D=2 \\ A=-2 \\ B=1 \end{cases} \Rightarrow A=-2, B=1, C=2, D=1$$

$$\begin{aligned} \text{thus } \int \frac{2x^2-2x+1}{x^4+x^2} dx &= \int \left(\frac{-2}{x} + \frac{1}{x^2} + \frac{2x+1}{x^2+1} \right) dx \\ &= \int \left(-\frac{2}{x} + \frac{1}{x^2} + \frac{2x}{x^2+1} + \frac{1}{x^2+1} \right) dx \\ &= \underline{-2 \ln|x| - \frac{1}{x} + \ln(x^2+1) + \arctan x + C} \end{aligned}$$

Problem 5. (20 points)

(a) Find the area of the region enclosed by the parabola $y = -x^2$ and the line $y = x - 2$.



$$\begin{cases} y = -x^2 \\ y = x - 2 \end{cases} \Rightarrow x^2 = x - 2 \Rightarrow x^2 + x - 2 = 0 \Rightarrow (x+2)(x-1) = 0$$

$$\Rightarrow x = -2, x = 1$$

$$\downarrow \quad \downarrow$$

$$y = -4 \quad y = -1$$

$$\text{Area} = \int_{-2}^1 (-x^2) - (x-2) dx$$

$$= \int_{-2}^1 -x^2 - x + 2 dx = \left[-\frac{x^3}{3} - \frac{x^2}{2} + 2x \right]_{x=-2}^{x=1}$$

$$= \left[-\frac{1}{3} - \frac{1}{2} + 2 \right] - \left[\frac{8}{3} - 2 - 4 \right]$$

$$= \frac{9}{2}$$

Trapezoidal

(b) Use the ~~Simpson's~~ rule with $n = 4$ to approximate $\int_1^2 \frac{1}{x} dx$ and also give an upper bound for the error of your estimate (without using the exact value of the integral).

$$a=1, b=2, n=4, \Delta x = \frac{2-1}{4} = \frac{1}{4}, f(x) = \frac{1}{x}$$

$$\begin{matrix} i & 0 & 1 & 2 & 3 & 4 \\ x_i & 1 & \frac{5}{4} & \frac{3}{2} & \frac{7}{4} & 2 \\ y_i & 1 & \frac{4}{5} & \frac{2}{3} & \frac{4}{7} & \frac{1}{2} \end{matrix}$$

$$T = \frac{\Delta x}{2} [y_0 + 2y_1 + 2y_2 + 2y_3 + y_4]$$

$$= \frac{1}{8} \left[1 + 2 \cdot \frac{4}{5} + 2 \cdot \frac{2}{3} + 2 \cdot \frac{4}{7} + \frac{1}{2} \right] = \frac{1}{8} \cdot \frac{1171}{210} = \frac{1171}{1680}$$

(≈ 0.6970)

$$f'(x) = -\frac{1}{x^2}, f''(x) = \frac{2}{x^3} \Rightarrow M_2 = \max_{1 \leq x \leq 2} |f''(x)|$$

$$= \max_{1 \leq x \leq 2} \frac{2}{x^3} = 2$$

$$\Rightarrow \left| T - \int_1^2 \frac{1}{x} dx \right| \leq \frac{M_2(b-a)^3}{12n^2} = \frac{2 \cdot 1^3}{12 \cdot 4^2} = \frac{1}{96} \quad (\approx 0.0104)$$